

Student ID:

Lab 04 Task

Your company, Make Abdullahpur Great Again (MAGA), is expanding its headquarters and requires a new network design to support various departments. The MAGA headquarters consists of the following departments across multiple floors: Engineering - Floor 1, Marketing - Floor 2, Sales - Floor 3, Administration - Floor 4, Management - Floor 5

Suppose they all share the prefix **192.(i).(i+3) % 10**. Below are the required devices for each of the departments:

$E(i) = (5i + 3) \bmod 41 + 5$	(Engineering)
$M(i) = (7i + 11) \bmod 26 + 5$	(Marketing)
$S(i) = (3i + 7) \bmod 36 + 5$	(Sales)
$A(i) = (11i + 17) \bmod 18 + 5$	(Administration)
$G(i) = (13i + 1) \bmod 9 + 5$	(Management)

where i is the last two digits your student ID

Implement and configure the topology, the devices should get their IP automatically from the DHCP server. Now for Engineering, Marketing and Sales the devices should use a separate DHCP server for automatic IP assignment & for Admin and Management they should use a router as a DHCP server. For each network you need to add minimum 4 end devices.

Lab 04 Grading Criteria

- 1. Correct Topology & Router Configuration (15 pts)**
 - Correct topology (5 pts)
 - Correct router configuration (10 pts)
- 2. DHCP Configuration (20 pts)**
 - Correct DHCP server configuration (10 pts)
 - Correct router as a DHCP server configuration (10 pts)
- 3. Routing (5 pts)**
 - Correct RIP configuration (5 pts)
- 4. Total: 40 pts**